

34721/34722 Linear Control Design 1

Spring Semester

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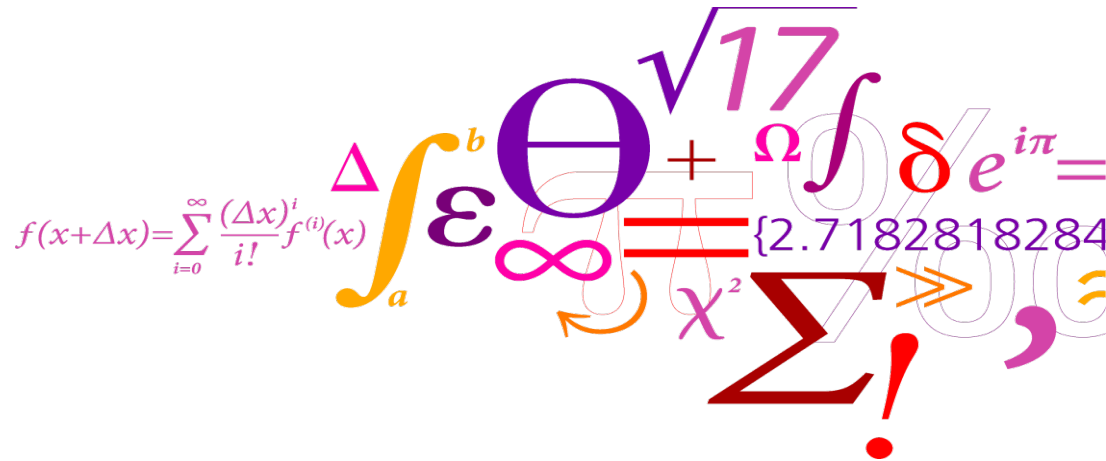


Lesson 10/13

Schedule (13:00-15:00)

.Unstable systems

.Intro to REGBOT balance



What you will learn today

After this lecture you will be able to

- Explain the concept of instability in dynamical systems
- Use Nyquist plots to stabilise open-loop unstable systems
- Explain the concept of cascaded control for REGBOT balance

Unstable systems

Unstable systems:

- Small perturbations remove system from equilibrium (e.g. inverted pendulum)
- Bounded input gives unbounded output (e.g. constant motor torque $\rightarrow$ infinite angular position)
- At least a pole in the RHP

$$G_1(s) = \frac{1}{s-1} \Leftrightarrow \dot{y} = y + u(t) \Rightarrow y(t) = y_0 e^t + \int_0^t e^{t-\tau} u(\tau) d\tau$$

- Multiple poles on the imaginary axis (on top of each other)

$$G_2(s) = \frac{1}{(s^2 + 1)^2} \xrightarrow{u(s)=1} y^{(4)} + \ddot{y} + y = 1\delta(t) \Rightarrow y(t) = 0.5 \sin(t) - 0.5t \cos(t)$$

$$\lim_{t \rightarrow \infty} |y(t)| = \infty$$

How can we stabilise an open-loop unstable system?

Stabilization of open-loop unstable systems

Consider the unstable system $G(s) = \frac{0.5(s+1)}{(0.1s+1)(-0.01s+1)} = -\frac{0.5s+0.5}{0.001s^2-0.09s-1}$

Can it be stabilized by a proportional controller K_{PS} ?

Algebraic approach:

$$G_{cl}(s) = \frac{K_{PS}G(s)}{1 + K_{PS}G(s)} = \frac{0.5K_{PS}(s+1)}{-0.001s^2 + (0.5K_{PS} + 0.09)s + 0.5K_{PS} + 1}$$

Must have roots only on the LHP

$\Delta = (0.5K_{PS} + 0.09)^2 + 4 \cdot 0.001(0.5K_{PS} + 1) \rightarrow$ always positive since $\Delta = 0$ only has complex roots

The poles of the closed-loop system are real:

$$p_{1,2} = 500 \left[(0.5K_{PS} + 0.09) \pm \sqrt{(0.5K_{PS} + 0.09)^2 + 0.004(0.5K_{PS} + 1)} \right]$$

We want $0.5K_{PS} + 0.09 < 0$ and $0.5K_{PS} + 1 < 0 \Rightarrow K_{PS} < -2$

Stabilization of open-loop unstable systems

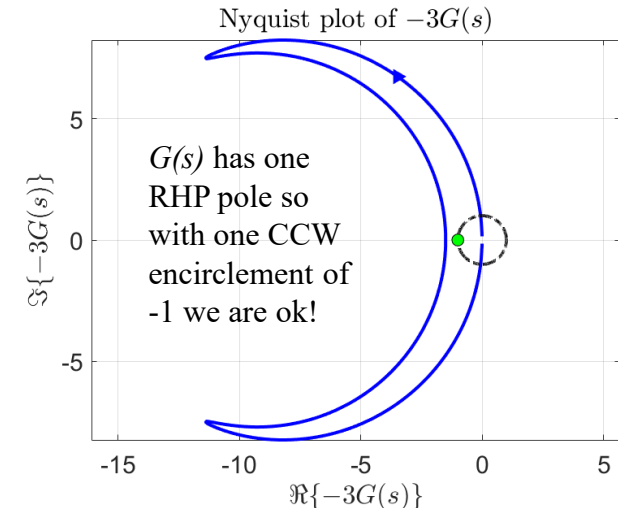
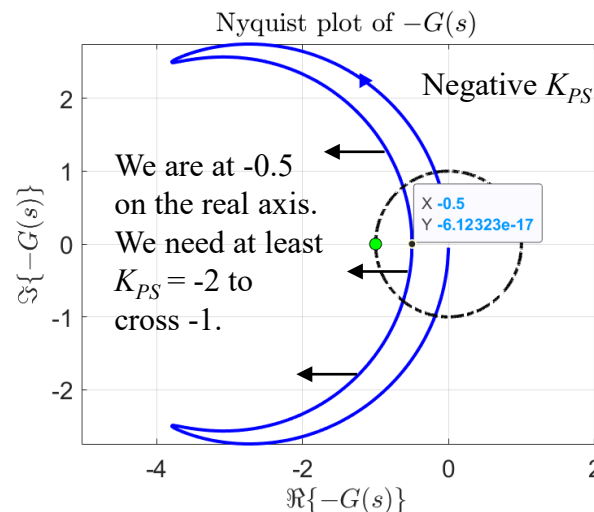
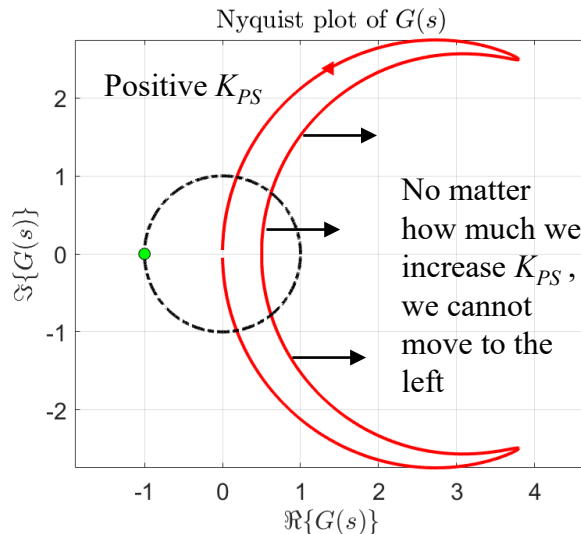
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Can it be stabilized by a proportional controller K_{PS} ?

Nyquist plot approach: $Z = N + P$

CW encirclements of (-1,0)
RHP poles of closed-loop
RHP poles of open-loop

For each RHP pole of G we need one CCW encirclement of $(-1,0)$



$$K_{PS} < -2$$

Stabilization of open-loop unstable systems

Example (Exam 2023): Consider a system described by a transfer function $G(s)$ with poles $\{0.045, -2, -3\}$ and open-loop Nyquist plot as shown in the figure. The plot intersects the real axis along the counter-clockwise direction at the point $(-0.741, 0)$. A P-controller K_{PS} is applied to the system and the loop is closed with unit feedback. The closed-loop system is stable for:

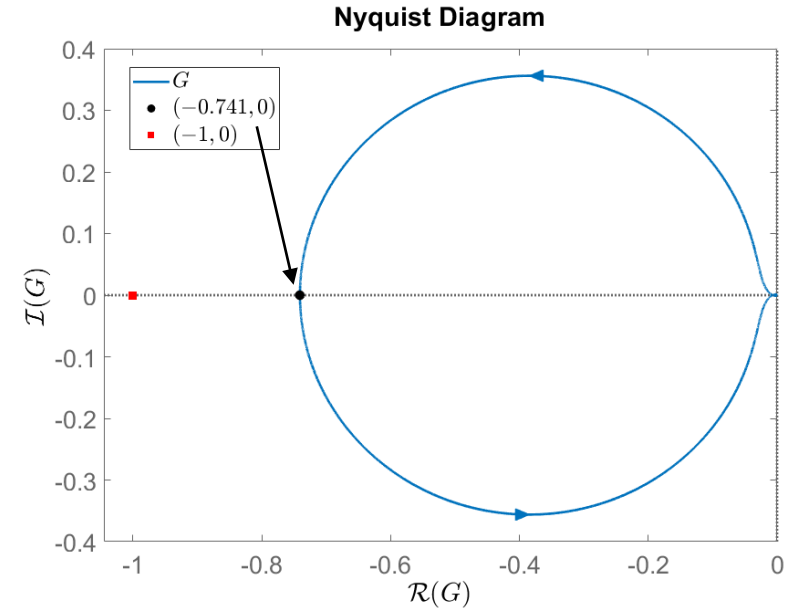
- a) $K_{PS} = 1.089$
- b) $K_{PS} = -0.741$
- c) $K_{PS} = 0.741$
- d) $K_{PS} = 1.5$
- e) $K_{PS} = 0.92$

For each RHP pole of G we need one CCW encirclement of $(-1, 0)$

CW encirclements of $(-1, 0)$

$$Z = N + P \Leftrightarrow 0 = N + 1 \Leftrightarrow N = -1$$

- The system is unstable because of the pole at 0.045.
- We need -1 CW encirclements or 1 CCW encirclement of $(-1, 0)$
- The graph direction is already CCW! We only need to multiply by an appropriate gain.
- How much should we multiply such that the intersection point is left of $(-1, 0)$?



Stabilization of open-loop unstable systems

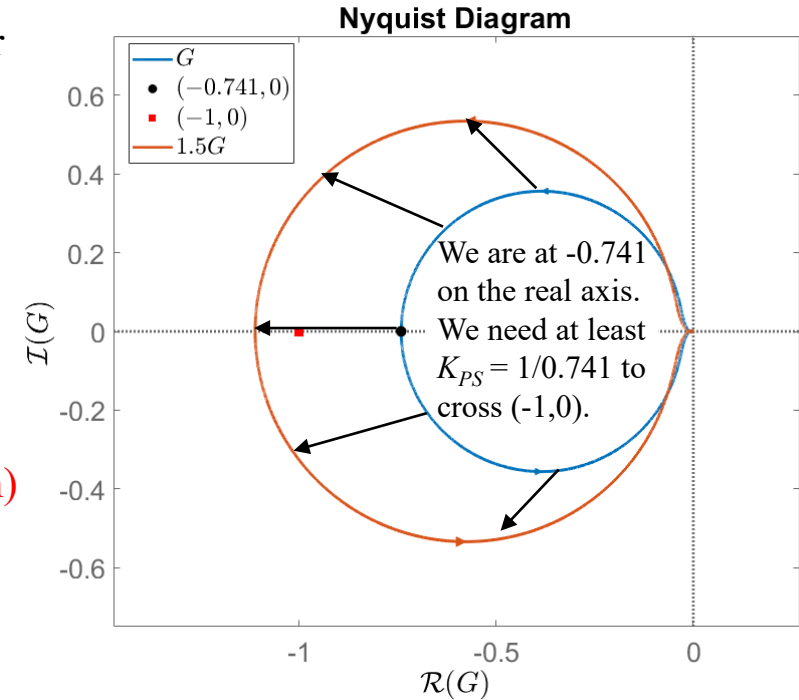
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- a) $K_{PS} = 1.089$ Too small gain (needs to be larger than 1.35)
- b) $K_{PS} = -0.741$ Negative gain (used coordinate of intersection)
- c) $K_{PS} = 0.741$ Too small gain (used current distance from 0)
- d) $K_{PS} = 1.5$ Correct (it is larger than 1.35)
- e) $K_{PS} = 0.92$ Too small gain (needs to be larger than 1.35)

- How much should we multiply such that the intersection point is left of $(-1, 0)$?

We are at -0.741 on the real axis. We need at least $K_{PS} = 1/0.741$ to cross exactly at $(-1, 0)$.

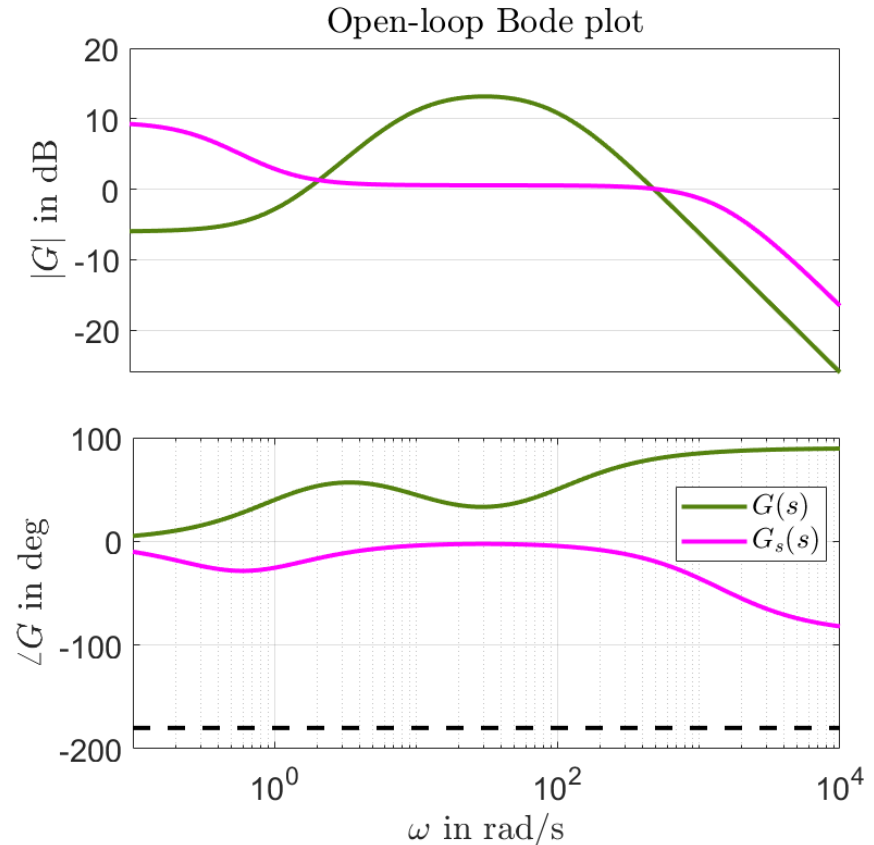
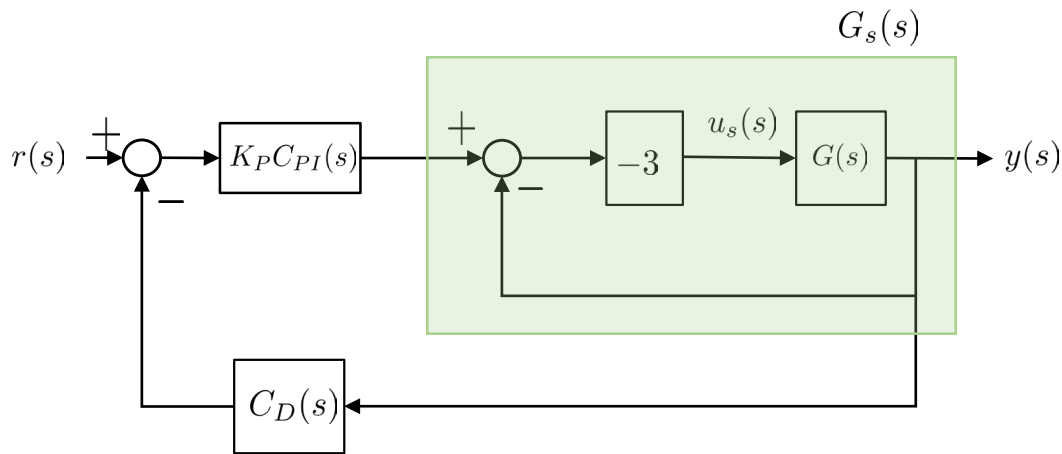
$$K_{PS} > \frac{1}{0.741} \approx 1.35$$



Stabilization & control of open-loop unstable systems

The stabilised system with $K_{PS} = -3$ has a new transfer function $G_s(s) = \frac{-3G(s)}{1 - 3G(s)} = \frac{1500s + 1500}{s^2 + 1410s + 500}$

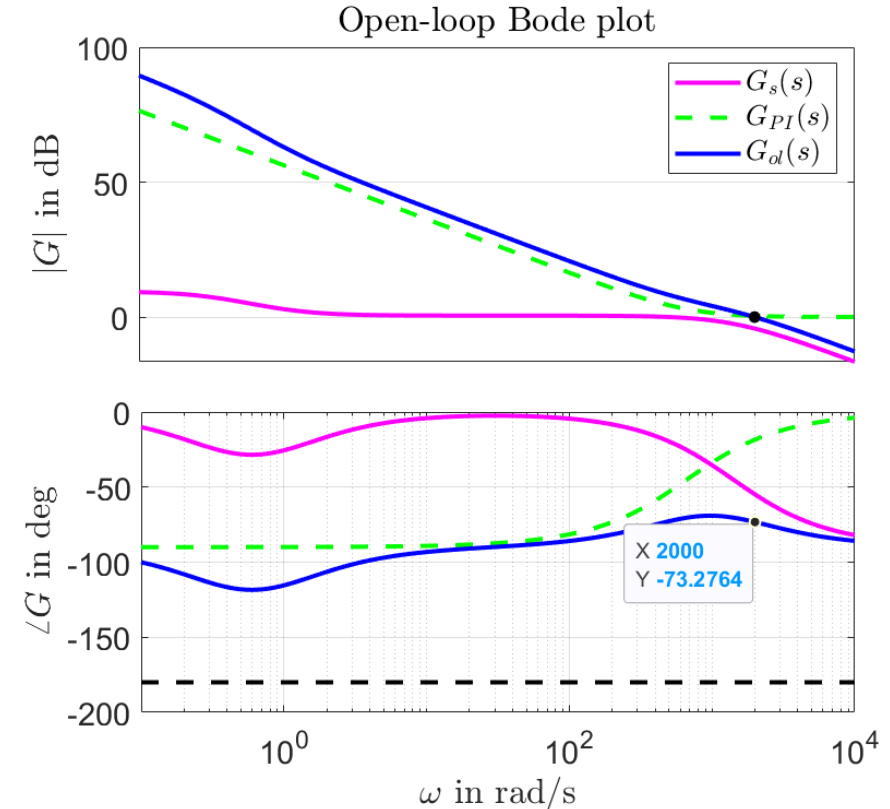
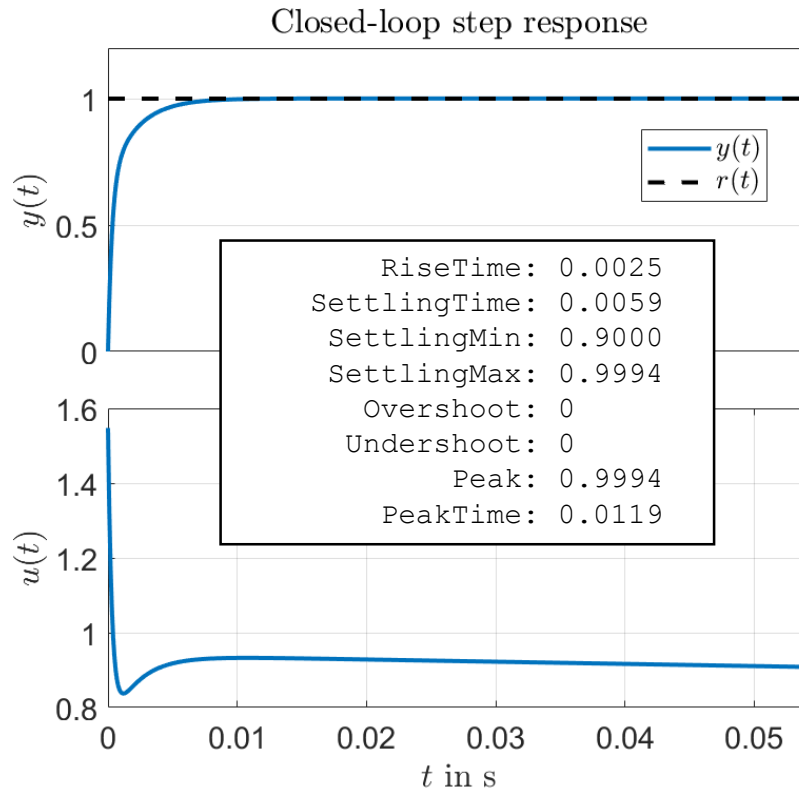
- No positive slopes in the magnitude curve (good response).
- Good phase margin $\rightarrow$ we do not need a Lead part.
- We want zero steady state error $\rightarrow$ PI.
- We want as high crossover frequency as possible to ensure good bandwidth.



Stabilization & control of open-loop unstable systems

The stabilised system with $K_{PS} = -3$ has a new transfer function $G_s(s) = \frac{-3G(s)}{1 - 3G(s)} = \frac{1500s + 1500}{s^2 + 1410s + 500}$

- Select $N_i = 3$, $\omega_c = 2000$ rad/s $\rightarrow \tau_i = \frac{N_i}{\omega_c} = 0.0015$, $K_P = 1.5475$



Stabilization & control of open-loop unstable systems

Alternative way:

We know that K_{PS} must be negative $\rightarrow$ include only the sign and define

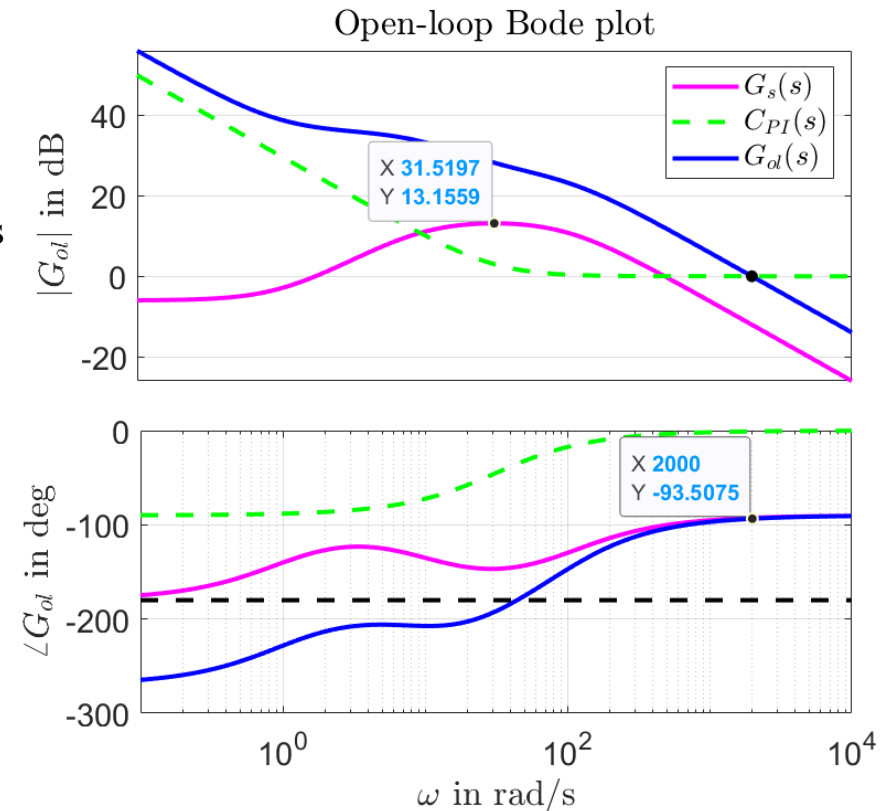
$$G_s(s) \triangleq -G(s) = \frac{0.5s + 0.5}{0.001s^2 - 0.09s - 1}$$

- From the Bode plot we can see that the magnitude curve of $G_s(s)$ is monotonically increasing until $\omega_i = 31.5$ rad/s $\rightarrow$ we want to avoid that $\rightarrow$ PI at that frequency

$$\tau_i = \frac{1}{\omega_i} = 0.0317 \Rightarrow C_{PI} = \frac{0.0317s + 1}{0.0317s}$$

- We select again $\omega_c = 2000$ rad/s (same bandwidth).
- We do not need Lead (already good phase margin $\sim 86^\circ$).

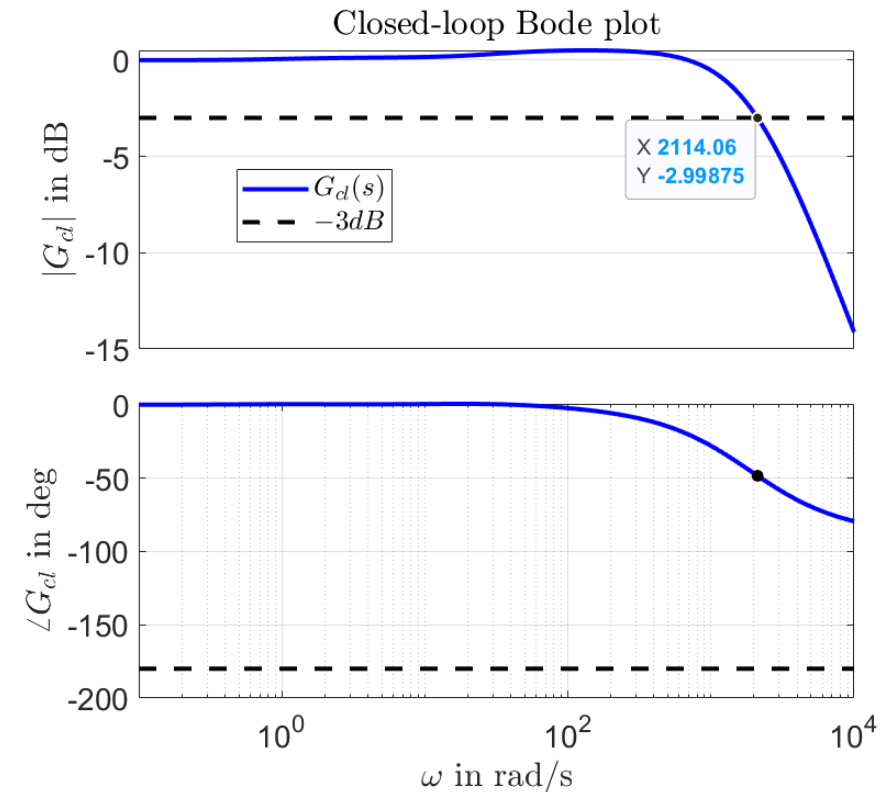
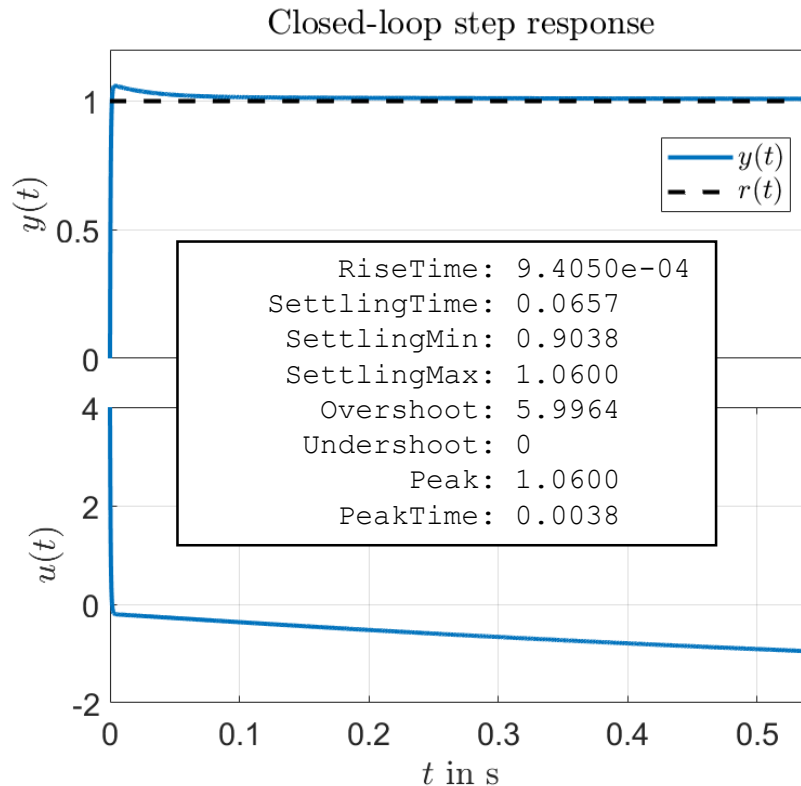
$$K_P = \frac{1}{|C_{PI}(s)G_s(s)|} \Big|_{s=j\omega_c} = 4$$



Stabilization & control of open-loop unstable systems

- Select $\omega_c = 2000$ rad/s, then $\tau_i = \frac{1}{\omega_i} = 0.0317$, $K_P = 4$

Stabilisation_open_loop_unstable.m



Stabilization & control of open-loop unstable systems

Stabilization of a given open-loop unstable system $G(s)$ – Method 1:

- **Step 1** : Find a gain K_{PS} that stabilizes $G(s)$ in unit feedback.
- **Step 2** : Close the loop and define the new system to be controlled as the closed-loop transfer function.
- **Step 3** : Select a crossover frequency (as large as possible for good bandwidth).
- **Step 4** : Design a feedback controller for the new system.
 - Check if a Lead part is needed for improving the phase margin.
 - Check if a PI part is needed for eliminating the steady-state error.
- **Step 5** : Test the closed-loop system performance.
 - Possibly iterate through the design, especially on the selection of the Lead-part and the crossover frequency ω_c .

Stabilization & control of open-loop unstable systems

Stabilization of a given open-loop unstable system $G(s)$ – Method 2:

- **Step 1** : Check if K_{PS} needs to be negative via Nyquist plot (as many counterclockwise encirclements of $(-1,0)$ as poles in the right half plane).
- **Step 2** : Set $G_S(s) = \text{sign}(K_{PS})G(s)$ and draw the Bode plot of $G_S(s)$.
 - Check whether an I-part is needed to remove increasing amplitude curves. If yes, place τ_i such that the amplitude curve is monotonically decreasing (to the extent that is possible).
 - Check whether a Lead-part can improve the phase margin and/or increase the crossover frequency ω_c without destroying the monotonically decreasing amplitude.
 - Select a crossover frequency and calculate τ_d and finally the K_P that gives the selected crossover frequency (remember the sign).
- **Step 3** : Test the closed-loop system performance.
 - Possibly iterate through the design, especially on the selection of the Lead-part and the crossover frequency ω_c (a lower ω_c and K_P do not always give a better result).

Introduction to Mandatory Assignment

REGBOT balance – Control architecture

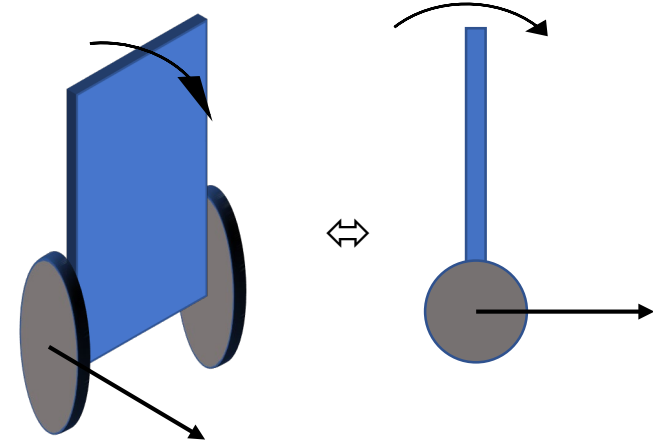
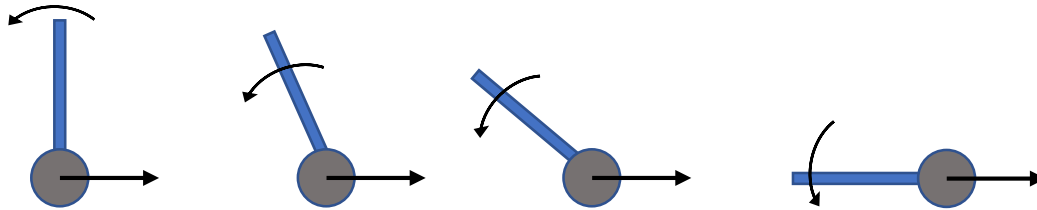
Contents

- How to get to balance
- How to get to balance at a given velocity
- How to get to balance at a given position
- Some tips

How to get to balance

Some facts:

- REGBOT $\Leftrightarrow$ Inverted pendulum with a cart.
- Constant speed cannot give balance.

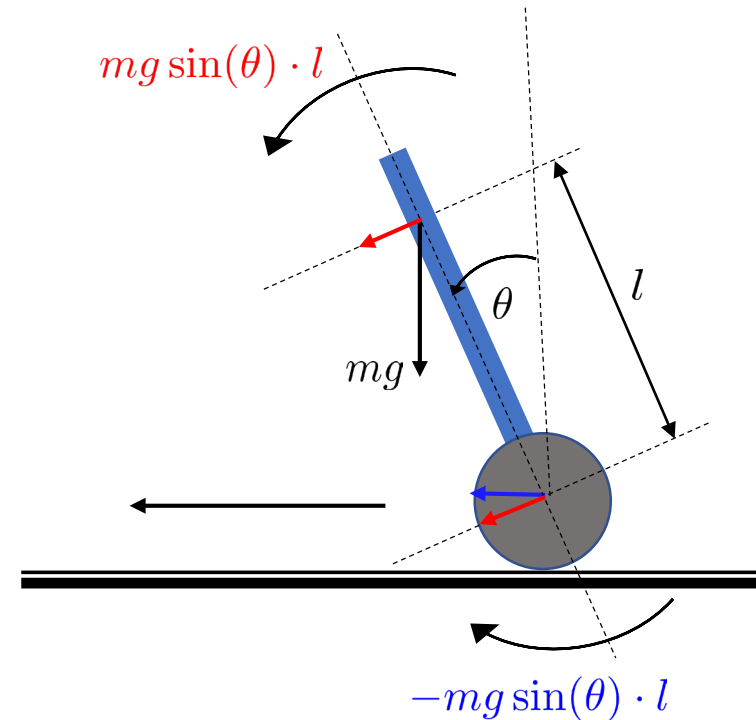


- REGBOT linear velocity = wheel radius $\times$ wheel angular speed = wheel radius $\times$ gear ratio $\times$ motor angular velocity
- We can achieve balance at some tilt angle with appropriate changing velocity command.

How to get to balance

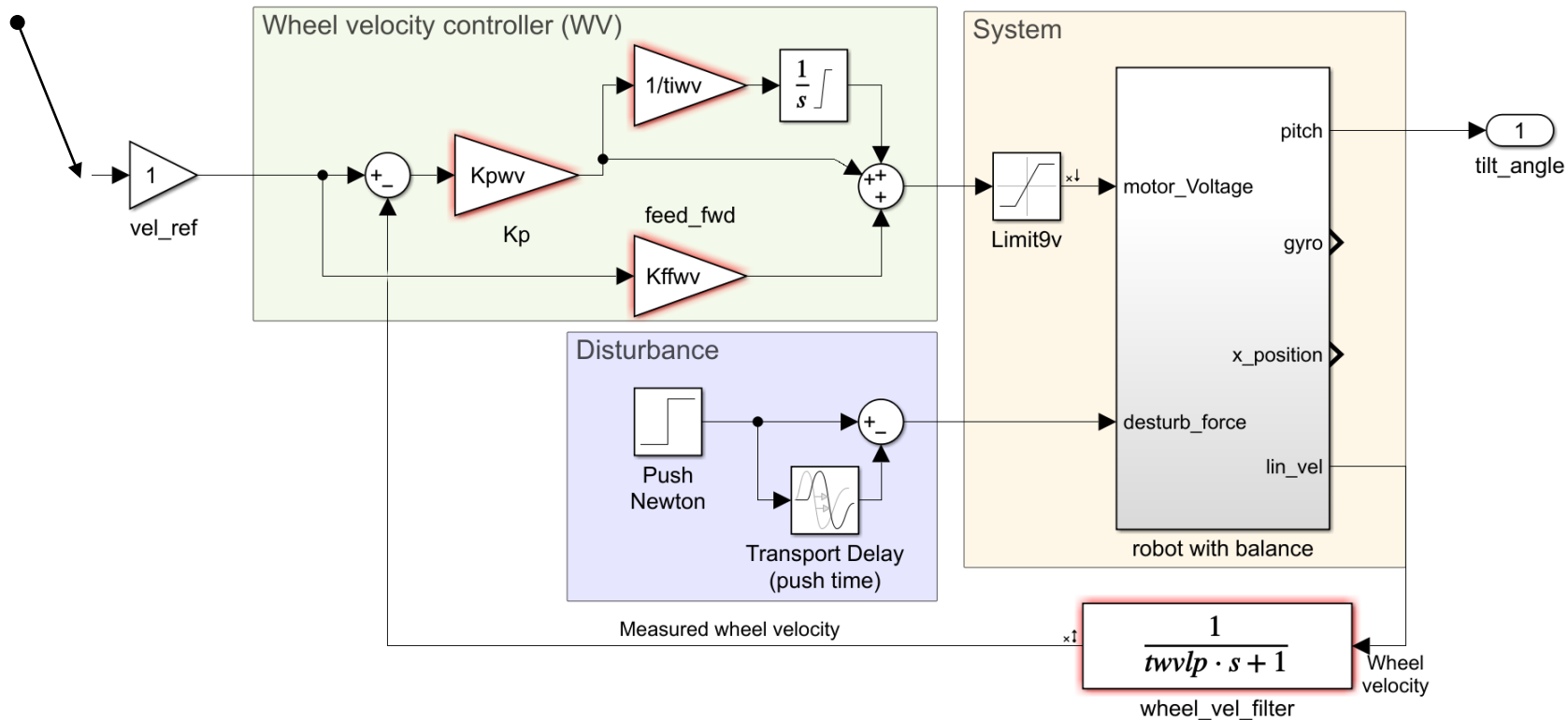
Some facts:

- Gravity has **component** that produces torque $\rightarrow$ tilt.
- We can counteract this **component** with a horizontal **force** in the same direction as the tilt.
- Force $\Leftrightarrow$ acceleration: A demand on the REGBOT linear speed (i.e. the wheel angular velocity) will produce an acceleration (force) command.
- We can achieve balance at some tilt angle with appropriate changing velocity command.



How to get to balance

Velocity demand such that it corresponds to the appropriate (for the balance) acceleration (i.e. force).

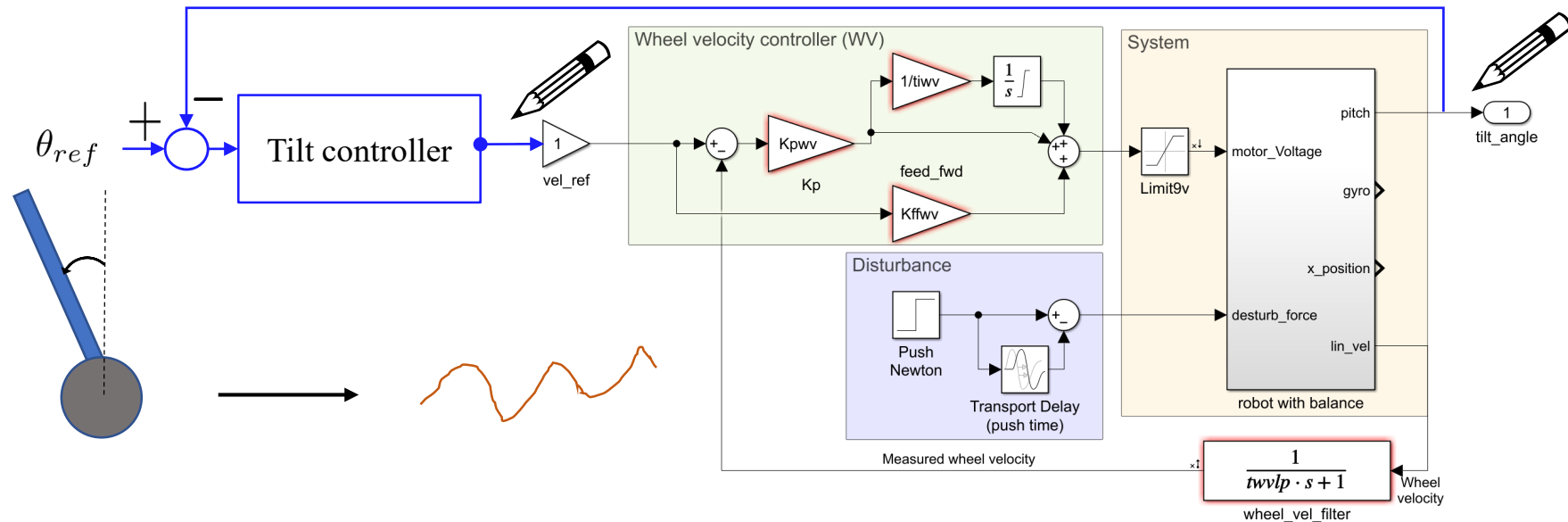


How to get to balance

Balance controller: Takes a desired tilt angle as reference (e.g. 3°) and outputs a special velocity demand (reference for the given velocity controller) such that it corresponds to the appropriate (for the balance) acceleration (i.e. force).

ACTION PLAN

1. Linearise  : From “vel_ref” to “tilt_angle”. This is your system transfer function.
2. Make a good controller for this system.
3. Close the loop in Simulink.

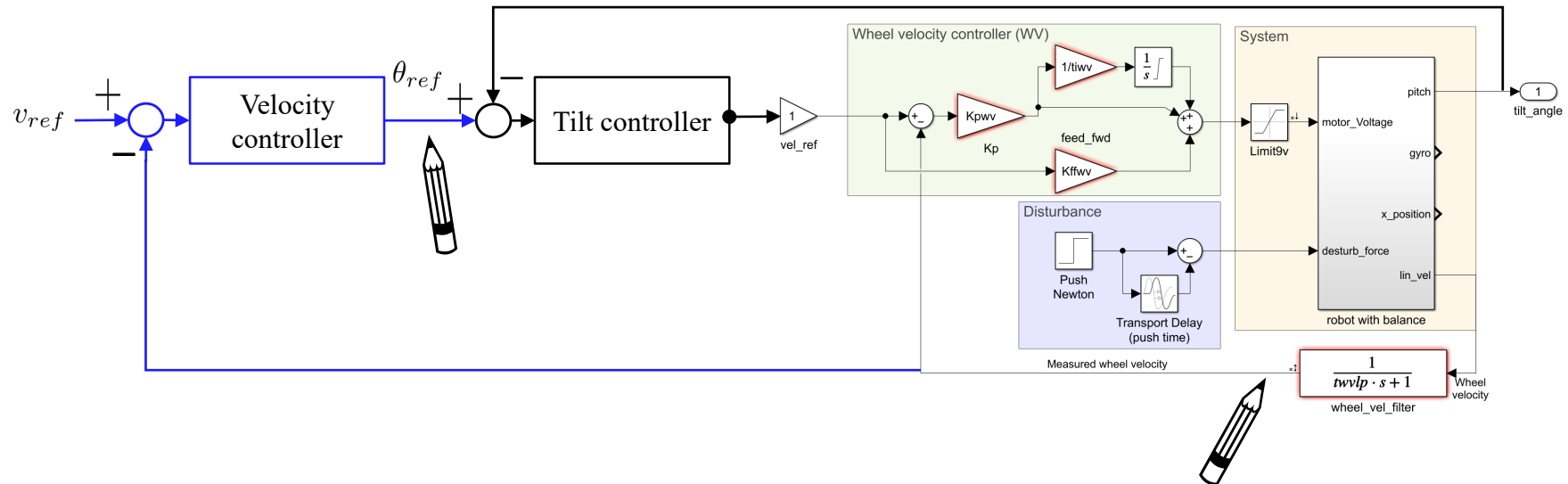


How to get to balance at a given velocity

Velocity controller: Takes a desired velocity as reference (e.g. 0.1 m/s) and outputs a special tilt angle demand (reference for the tilt controller) such that it forces the REGBOT move with the given velocity (e.g. tilt forward will make the robot move forward in order to maintain balance as well).

ACTION PLAN

1. Linearise  : From “ θ_{ref} ” to output of the “wheel_vel_filter”. This is your new system transfer function.
2. Make a good controller for this system.
3. Close the loop in Simulink.

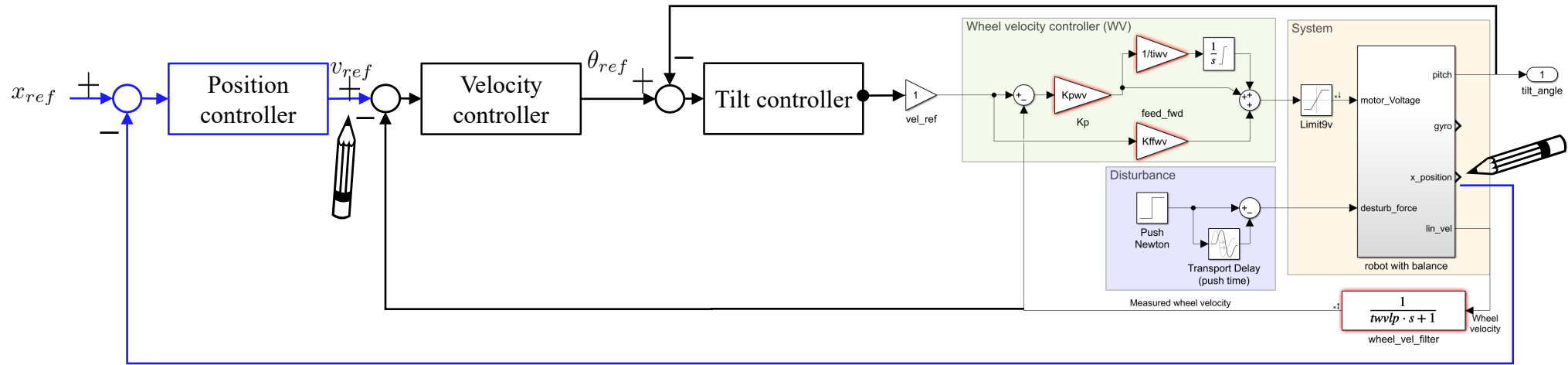


How to get to balance at a given position transit

Position controller: Takes a desired positions reference (e.g. 2 m/s) and outputs a special velocity demand (reference for the velocity controller) such that it forces the REGBOT move to the given position (e.g. tilt forward until it almost reaches the target and then tilt backwards to start maintaining the balance).

ACTION PLAN

1. Linearise  : From “v_ref” to “x_position”. This is your new system transfer function.
2. Make a good controller for this system.
3. Close the loop in Simulink.

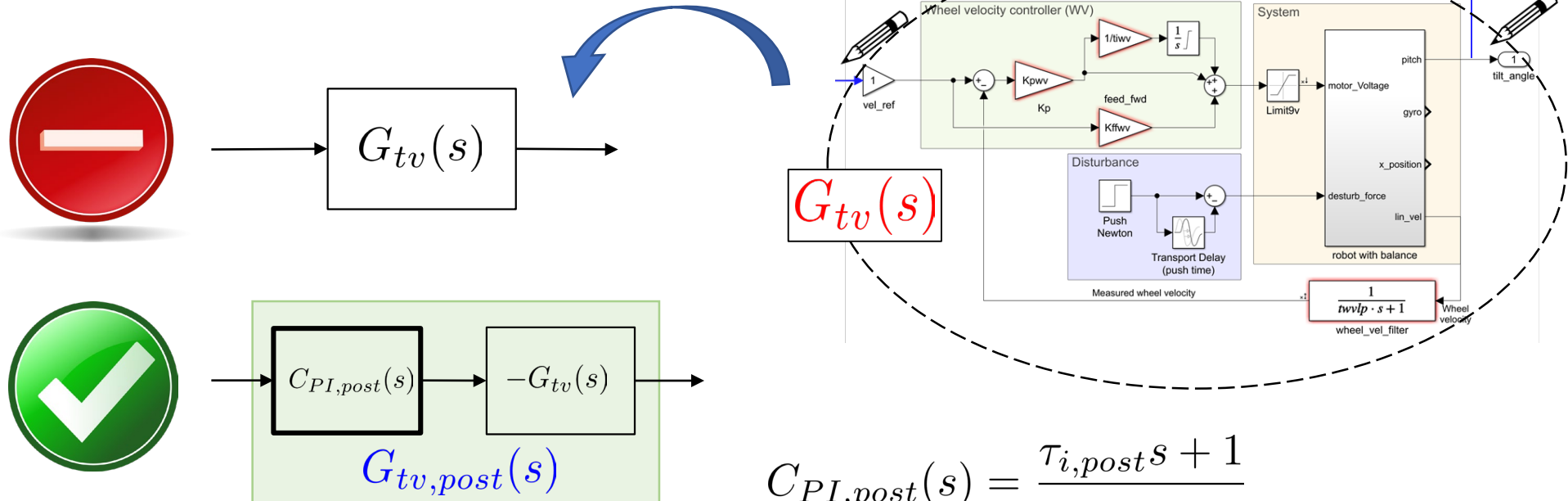


Some tips

Tilt angle (balance) controller:

- The transfer function $G_{tv}(s)$ you get from the first linearization is **unstable**!
- You will need one more PI block $C_{PI_post}(s)$ in open loop to stabilise it. This PI-part is also referred to as “post-integrator”. The new transfer function to be controlled is

$$G_{tv,post}(s) = -C_{PI,post}(s)G_{tv}(s)$$



$$C_{PI,post}(s) = \frac{\tau_{i,post}s + 1}{\tau_{i,post}s}$$

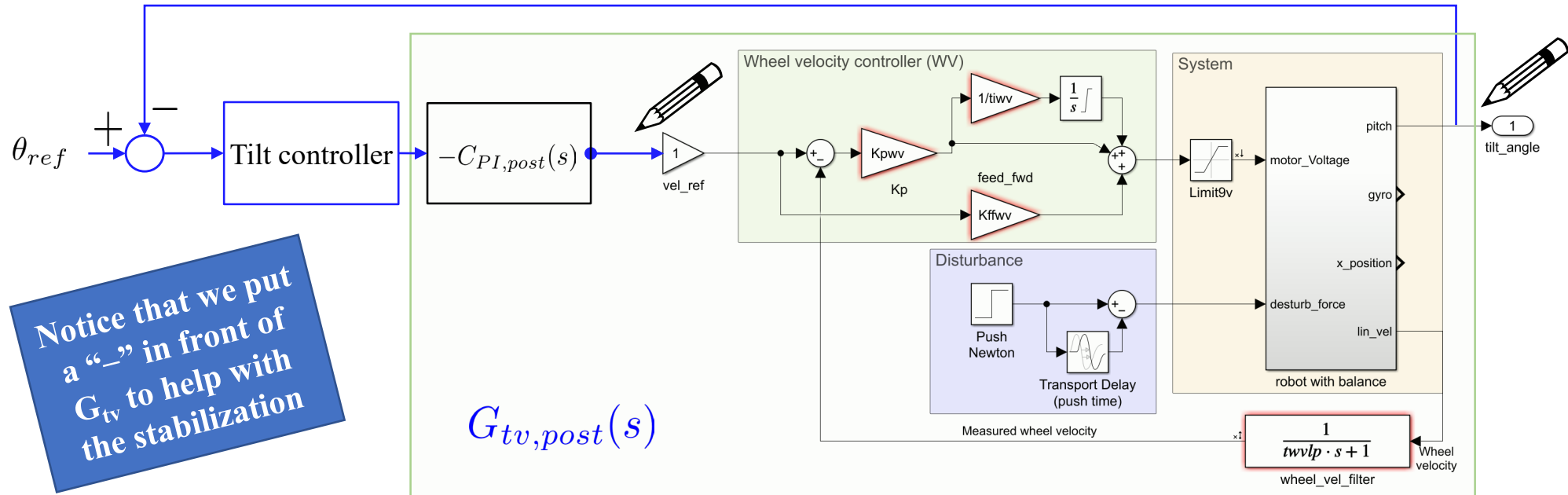
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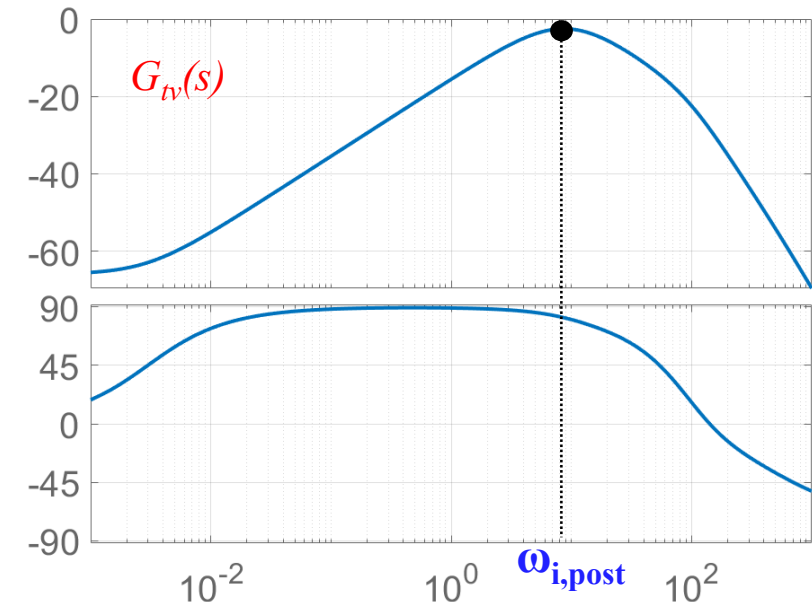
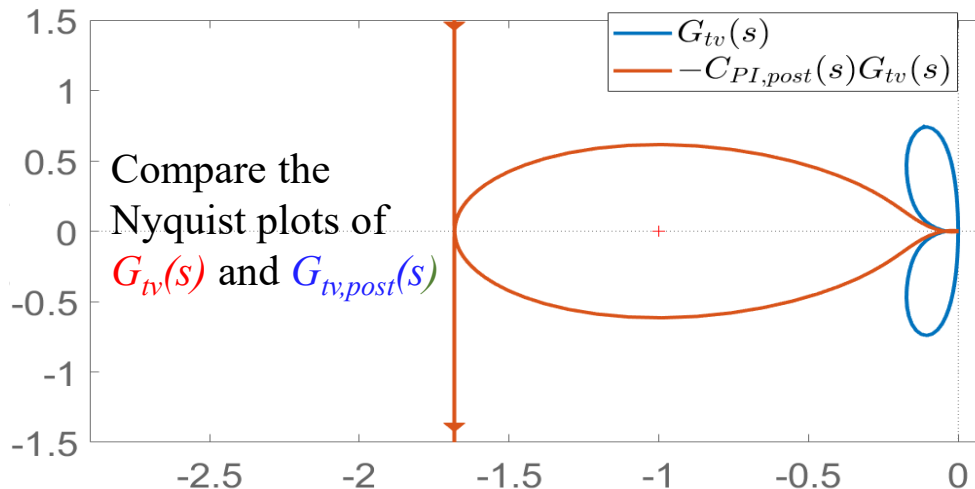
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$$G_{tv,post}(s) = -C_{PI,post}(s)G_{tv}(s) \qquad C_{PI,post}(s) = \frac{\tau_{i,post}s + 1}{\tau_{i,post}s}$$

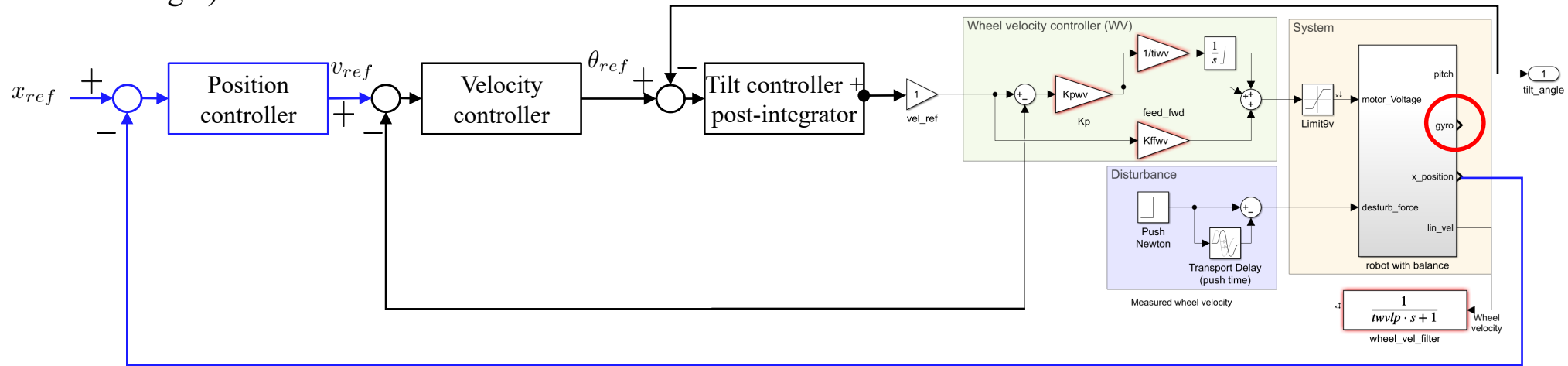
- To find the post-integrator time constant $\tau_{i,post}$,
 - ✓ Make a Bode plot of $G_{tv}(s)$
 - ✓ Find the location $\omega_{i,post}$ of the magnitude peak
 - ✓ Select $\tau_{i,post} = 1/\omega_{i,post}$



Some tips

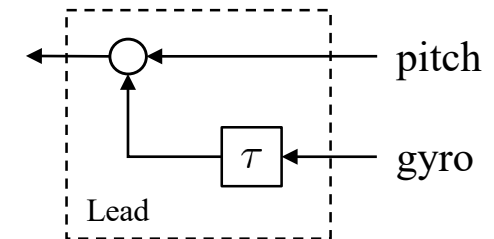
Tilt angle (balance) controller:

- For the Lead part of the tilt angle controller you can use the gyro measurement (derivative of tilt angle).



$$\frac{\tau s + 1}{\alpha \tau s + 1} \theta = \underbrace{\frac{1}{\alpha \tau s + 1}}_{\text{low-pass filter}} \left(\tau \underbrace{s\theta}_{\dot{\theta}=\text{gyro}} + \theta \right) \rightarrow \tau \cdot \text{gyro} + \theta$$

Not needed



Next lecture 11

Limited systems

- Explain rate and amplitude limitations in systems
- Use mitigation strategies in the control design of limited systems

Today's Exercises

Group work: REGBOT balance (mandatory assignment)

Home alone: Continue on the theoretical exercises LCD1: 1, 2, 4, 5, 6, 7, 9